

Question				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
		(iii)		as the main carbon chain length increases, the boiling temperature increases (or converse) (1) award (1) for either of following lengthening the chain increases the Van der Waals forces between molecules more heat energy is required to overcome the increased forces (or converse)		1	1	2		
	(b)	(i)		$C_6H_{14} + 9\frac{1}{2}O_2 \rightarrow 6CO_2 + 7H_2O$ accept $2C_6H_{14} + 19O_2 \rightarrow 12CO_2 + 14H_2O$ formulae (1) balancing (1) – only awarded if all formulae are correct		2		2		
		(ii)		the number and types of bond are the same (1) the energy absorbed by breaking bonds and the energy released by forming bonds will be similar (1)		2		2		
		(iii)		it has the lowest boiling point / is the most volatile / vaporises most quickly accept explanation of lower boiling point		1		1		
		(iv)	I	energy absorbed = $(5 \times 348) + (14 \times 413) + (6.5 \times 495) = 10739.5$ (1) energy released = $(6 \times 1072) + (14 \times 464) = 12928$ (1) enthalpy change = $10739.5 - 12928 = -2188.5$ (1)		3		3	2	